Malaysian Banking Analytics: A Synthetic SQL Project

Project Report

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Tool: MySQL 8.0 Workbench

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Abstract

This report documents a self-initiated SQL project built entirely from scratch. The project involved designing a relational database, generating synthetic Malaysian banking data manually in SQL, and writing 14 business analysis queries to extract meaningful insights. The goal was to demonstrate practical skills in database design, data engineering, and business analytics using SQL as the primary tool.

Part 1: Database Design and Synthetic Data Generation

1. Project Overview

This project simulates a fictional Malaysian retail bank called **mybank_analytics**. Every piece of data in this project was written and generated by me directly in SQL. There was no external dataset used, no CSV import, and no automated data generation tool. Each row was crafted manually to reflect realistic Malaysian banking scenarios, naming conventions, income levels, and financial products.

The motivation behind building synthetic data rather than using a public dataset was straightforward. It forces a deeper understanding of data relationships, business logic, and domain knowledge. When you generate the data yourself, you understand exactly what each field means and why each relationship exists.

2. Database Architecture

The database was built in three separate SQL files to keep the work organised and maintainable.

- **File 1 (01_schema_setup.sql):** All table definitions, constraints, indexes, views, and the stored procedure
- **File 2 (02_synthetic_data.sql):** All INSERT statements populating the database
- **File 3 (03_analysis_queries.sql):** All 14 business analysis queries

The database runs on MySQL 8.0 with the utf8mb4 character set and utf8mb4_unicode_ci collation, which supports multilingual characters including Malay and Tamil names.

3. Schema Design

The schema consists of eight relational tables. Each table was designed with a clear business purpose.

Table 1: *Data Description*

Table	Purpose	Rows
branches	10 fictional bank branches across Malaysia	10
customers	80 fictional Malaysian customers	80
accounts	Savings, current, and fixed deposit accounts	120+
transactions	Banking transaction records	600+
loans	Personal, home, car, and SME loans	50
loan_repayments	Monthly instalment payment records	150
cards	Debit and credit card records	70
card_transactions	Merchant spending records	200+

Total synthetic rows generated: approximately 1,650+

3.1 Table Design Decisions

Each table was designed with careful attention to constraints and data integrity.

Primary Keys were defined using AUTO_INCREMENT integers on all eight tables. Named constraints (for example, pk_branches, pk_customers) were used instead of unnamed ones for clarity and maintainability.

Foreign Keys link all child tables back to their parent tables. For example, accounts references both customers and branches. The ON DELETE RESTRICT and ON UPDATE CASCADE rules were set deliberately so that deleting a customer is blocked if they still have active accounts or loans.

Unique Constraints enforce business rules. The customers table enforces unique NRIC numbers (uq_nric) and unique email addresses (uq_email). The accounts table enforces a unique account number format (uq_account_number) using the Malaysian-style format MY-XXXXXX-XX.

CHECK Constraints validate business logic at the database level. These include checks that balance >= 0, amount > 0, monthly_income > 0, principal_amount > 0, and tenure_months > 0. This means invalid data is rejected before it even reaches the application layer.

ENUM Types were used extensively to restrict field values to valid options only. Examples include account_type ENUM('savings','current','fixed_deposit'), income_tier

ENUM('B40','M40','T20'), channel ENUM('ATM','online','counter','mobile'), and loan_status ENUM('active','settled','defaulted','npl'). This eliminates the possibility of free-text errors in categorical fields.

3.2 Indexes

A total of 19 indexes were created across frequently queried columns. These cover fields that appear regularly in WHERE clauses, JOIN conditions, and GROUP BY operations. Examples include idx_transactions_date, idx_loans_status, idx_customers_income_tier, and idx_card_txn_category. These indexes ensure that the analytical queries in Part Two execute efficiently even as the dataset scales.

4. Views

Four analytical views were created to pre-compute commonly needed joins and aggregations. These act as reusable query building blocks.

vw_account_summary flattens the accounts table with customer and branch details into a single readable view. It is useful for basic account-level reporting without writing repetitive joins.

vw_loan_portfolio joins loans with customer demographics and branch information. It also includes a computed Debt Service Ratio (DSR) field calculated as monthly_instalment / monthly_income * 100. This gives a direct indication of how much of a customer's income goes toward loan repayment.

vw_branch_kpi is the most complex view. It aggregates customer count, account count, total deposits, total loan book, and outstanding loan balance per branch. It also computes the Loan-to-Deposit Ratio (LDR) using SUM(outstanding_balance) / NULLIF(SUM(balance), 0) * 100. The NULLIF function prevents division-by-zero errors for branches with no deposits.

vw_overdue_loans flags all loan repayments where the status is missed or partial and the due date has already passed. It uses DATEDIFF(CURDATE(), lr.due_date) to compute the number of days overdue dynamically, meaning the view stays current without any data refresh.

5. Stored Procedure

A stored procedure named GetCustomerPortfolio was written to return the full financial profile of any customer in a single database call. It accepts a customer_id as an input parameter and

returns four result sets: the customer's personal profile, all active accounts, all loans, and all active cards. The card number is masked using CONCAT('****', card_number_last4) to follow security best practices. This procedure demonstrates the ability to write parameterised, multi-result-set stored procedures in MySQL.

6. Synthetic Data Design

Generating realistic synthetic data required careful planning across several dimensions.

Customer data includes 80 fictional Malaysian individuals spread across all three Malaysian income tiers: B40 (bottom 40%), M40 (middle 40%), and T20 (top 20%). Names were chosen to reflect Malaysia's multiracial population, covering Malay, Chinese, and Indian naming conventions. Monthly income values are calibrated to each tier. NRIC numbers follow the real Malaysian format YYMMDD-SS-NNNN but are entirely fictional and non-identifiable.

Account data covers 120+ accounts across three types. Each customer holds between one and three accounts. Fixed deposit accounts have realistic maturity dates set one to three years after the opening date, with interest rates ranging from 3.75% to 4.50%. Savings accounts use rates between 1.00% and 2.00%. Three accounts were intentionally set to dormant status to represent inactive accounts, as this is a common scenario in real banking.

Transaction data covers 600+ records spanning January to mid-2023. Transactions include salary credits, ATM withdrawals, mobile bill payments, IBG transfers, DuitNow transfers, FPX credits, and fixed deposit interest payouts. Malaysian utility providers (TNB, Unifi TM, Maxis, Celcom, Astro) and payment references (PTPTN, IWK) were used to keep the context authentic.

Loan data covers 50 loans across four product types. Home loan tenures range from 180 to 360 months. SME loans carry the highest interest rates (5.50% to 6.20%). Several loans are marked as settled to represent fully repaid accounts. Monthly instalments were calculated using standard amortisation logic to ensure they are mathematically consistent with the principal, rate, and tenure.

Loan repayment data includes three repayment records per loan. Some repayments are intentionally set as missed or partial with no paid_date, simulating real delinquency scenarios. This feeds directly into the credit risk analysis in Part Two.

Card and card transaction data covers 70 cards split across debit (Visa/Mastercard) and credit cards. Credit card limits range from MYR 15,000 to MYR 50,000. Card transaction merchants are real Malaysian businesses including Jaya Grocer, Aeon, Tesco, McDonald's, Nando's, Grab, AirAsia, Malaysia Airlines, Shopee, Lazada, Pantai Hospital, and KPJ Hospital.

Part 2: Business Analysis and Findings

1. Analysis Approach

The 14 analysis queries were organised around five business themes: branch performance, customer profiling, credit risk, transaction behaviour, and card spending. Advanced SQL techniques were used throughout, including window functions (RANK(), NTILE(), LAG(), SUM() OVER()), Common Table Expressions (CTEs), multi-level JOINs, conditional aggregation with CASE WHEN, and the NULLIF function for safe division. The analysis queries were written in 03_analysis_queries.sql and executed against the populated mybank_analytics database.

2. Branch Deposit Distribution (Q1)

The Johor Bahru Branch holds the largest deposit base at MYR 2,260,600 across 30 accounts, followed by the KLCC Branch in Kuala Lumpur at MYR 1,618,600. Together these two branches account for roughly 62% of the bank's total deposits. The Subang Jaya Branch has the lowest average balance per account at MYR 18,440, suggesting a younger and lower-income customer profile. Three branches (Kuching, Kota Kinabalu, and Melaka) show zero activity and are treated as newly established branches pending operational rollout.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	full_name	income_tier	occupation	branch_name	total_accounts	total_balance_myr	monthly_income_myr	wealth_to_income_ratio
8	Zulkifli Omar	T20	Business Owner	Johor Bahru Branch	2	320,000.00	18,500.00	17.3
23	Noraini Hamid	T20	Doctor	KLCC Branch	2	275,000.00	16,800.00	16.4
68	Vikram Balaji	T20	Architect	Johor Bahru Branch	2	270,000.00	13,700.00	19.7
38	Wong Kok Wai	T20	Doctor	KLCC Branch	2	270,000.00	18,500.00	14.6
76	Deepa Mohan	T20	Software Engineer	Johor Bahru Branch	2	263,000.00	11,000.00	23.9
58	Chan Wai Keng	T20	Doctor	Georgetown Branch	2	240,000.00	17,600.00	13.6
3	Abdullah Sani	T20	Doctor	KLCC Branch	2	235,000.00	15,200.00	15.5
33	Aisyah Razak	T20	Architect	KLCC Branch	2	228,000.00	12,200.00	18.7
72	Kavitha Raj	T20	IT Manager	Johor Bahru Branch	3	225,600.00	11,600.00	19.4
52	Lim Bee Geok	T20	IT Manager	Johor Bahru Branch	2	223,000.00	11,300.00	19.7

Figure 1: *Result of Query 1*

3. Top Customer Profiling (Q2)

The top 10 customers by total balance are all from the T20 income tier. Their combined deposits exceed MYR 2.6 million. Dominant occupations include doctors, architects, IT managers, and business owners. Five of the top 10 are based at the Johor Bahru Branch. The wealth-to-income ratio, calculated as total balance divided by monthly income, ranges from 13.6x to 23.9x. Deepa

Mohan records the highest ratio at 23.9x, indicating strong savings behaviour relative to income. These customers are well-suited for private banking and investment product offerings.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

branch_name	state	total_accounts	savings_accounts	current_accounts	fd_accounts	total_deposits_myr	avg_balance_myr
Johor Bahru Branch	Johor	30	12	11	7	2,260,600.00	75,353.33
KLCC Branch	Kuala Lumpur	26	11	7	8	1,618,600.00	62,253.85
Petaling Jaya Branch	Selangor	19	11	4	4	708,300.00	37,278.95
Georgetown Branch	Penang	16	10	3	3	630,800.00	42,053.33
Shah Alam Branch	Selangor	12	5	4	3	531,500.00	44,291.67
Ipoh Branch	Perak	14	7	5	2	485,800.00	34,700.00
Subang Jaya Branch	Selangor	12	10	2	0	184,400.00	18,440.00
Kuching Branch	Sarawak	0	0	0	0	0.00	NULL
Kota Kinabalu Branch	Sabah	0	0	0	0	0.00	NULL
Melaka Branch	Melaka	0	0	0	0	0.00	NULL

Figure 2: *Result of Query 2*

4. Loan Portfolio Analysis (Q3)

Home loans dominate the portfolio with MYR 6.43 million in total principal across 14 active loans. The outstanding ratio for home loans stands at 87.53%, confirming these are recently disbursed long-tenure loans. SME loans carry the highest interest rate among secured products at 5.80% average. Personal loans, though smaller in volume (MYR 560,000 total principal), have the highest rate at 7.96% due to being unsecured. Car loans are the most affordable at 3.12% average. Three home loans, three personal loans, and one car loan have been fully settled.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

loan_type	loan_status	total_loans	total_principal_myr	total_outstanding_myr	avg_interest_rate_pct	avg_tenure_months	outstanding_ratio_pct
personal	active	14	560,000.00	402,950.00	7.96	45	71.96
personal	settled	3	155,000.00	0.00	7.57	36	0.00
home	active	14	6,430,000.00	5,628,400.00	4.15	296	87.53
home	settled	2	860,000.00	0.00	4.25	270	0.00
car	active	11	1,291,000.00	976,550.00	3.12	81	75.64
car	settled	1	55,000.00	0.00	3.00	60	0.00
sme	active	5	2,070,000.00	1,807,000.00	5.80	67	87.29

Figure 3: *Result of Query 3*

5. Transaction Trend Analysis (Q4 and Q13)

January 2023 recorded 99 transactions with a total value of MYR 4.82 million. This elevated figure reflects the initial account funding phase where salary credits and account opening deposits were processed together. February 2023 shows the most diverse transaction mix, with deposits, withdrawals, transfers, and bill payments all active. The month-over-month analysis using LAG() shows a count growth of +9.09% from January to February, but a sharp value decline of 67.95%, which means more transactions happened but they were smaller in size.

Activity drops significantly from March onward, which reflects the synthetic data boundary rather than a real decline.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

txn_month	transaction_type	txn_count	total_amount_myr	avg_amount_myr	max_amount_myr
2023-01	deposit	97	4,816,100.00	49,650.52	200,000.00
2023-01	withdrawal	1	800.00	800.00	800.00
2023-01	payment	1	150.00	150.00	150.00
2023-02	deposit	23	1,014,100.00	44,091.30	145,000.00
2023-02	withdrawal	20	34,800.00	1,740.00	4,000.00
2023-02	transfer_in	10	105,000.00	10,500.00	20,000.00
2023-02	transfer_out	30	385,000.00	12,833.33	25,000.00
2023-02	payment	25	4,794.00	191.76	350.00
2023-03	deposit	29	855,700.00	29,506.90	130,000.00
2023-03	withdrawal	4	8,100.00	2,025.00	5,000.00
2023-03	transfer_in	1	10,000.00	10,000.00	10,000.00
2023-03	transfer_out	3	12,000.00	4,000.00	5,000.00
2023-03	payment	4	880.00	220.00	320.00
2023-04	deposit	26	865,000.00	33,269.23	118,000.00
2023-04	interest	18	61,913.96	3,439.66	6,750.00
2023-05	interest	6	15,753.13	2,625.52	4,706.25
2023-06	interest	2	3,000.00	1,500.00	1,800.00

Figure 4: *Result of Query 4*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

txn_month	txn_count	total_amount_myr	prev_month_count	prev_month_amount	mom_count_growth_pct	mom_value_growth_pct
2023-01	99	4,817,050.00	NULL	NULL	NULL	NULL
2023-02	108	1,543,694.00	99	4817050.00	9.09	-67.95
2023-03	41	886,680.00	108	1543694.00	-62.04	-42.56
2023-04	44	926,913.96	41	886680.00	7.32	4.54
2023-05	6	15,753.13	44	926913.96	-86.36	-98.30
2023-06	2	3,000.00	6	15753.13	-66.67	-80.96

Figure 5: *Result of Query 13*

6. Branch Risk and Loan-to-Deposit Ratio (Q5, Q7, Q8)

This is the most critical finding across all 14 queries. Every active branch in the database exceeds the Bank Negara Malaysia (BNM) recommended Loan-to-Deposit Ratio ceiling of 90%.

Table 2: *Loan-to-Deposit Ratio*

Branch	LDR (%)	Status
Subang Jaya	438.26%	Over-lent
Georgetown	363.76%	Over-lent
Johor Bahru	281.58%	Over-lent
Petaling Jaya	218.98%	Over-lent
KLCC	205.54%	Over-lent
Shah Alam	123.80%	Over-lent
Ipoh	92.22%	Over-lent

The Subang Jaya Branch is in the most concerning position, with four times more loans outstanding than deposits on hand. Even the best-performing branch, Ipoh, marginally breaches the 90% threshold. This scenario was intentionally designed to simulate a realistic liquidity stress environment. In a real bank, these branches would trigger mandatory intervention from the risk and treasury departments.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

branch_name	state	total_loans	active_loans	defaulted_loans	npl_loans	settled_loans	default_rate_pct	npl_exposure_myr
KLCC Branch	Kuala Lumpur	13	12	0	0	1	0.00	0.00
Petaling Jaya Branch	Selangor	8	5	0	0	3	0.00	0.00
Georgetown Branch	Penang	6	5	0	0	1	0.00	0.00
Johor Bahru Branch	Johor	13	13	0	0	0	0.00	0.00
Ipoh Branch	Perak	2	1	0	0	1	0.00	0.00
Shah Alam Branch	Selangor	4	4	0	0	0	0.00	0.00
Subang Jaya Branch	Selangor	4	4	0	0	0	0.00	0.00
Kuching Branch	Sarawak	0	0	0	0	0	NULL	0.00
Kota Kinabalu Branch	Sabah	0	0	0	0	0	NULL	0.00
Melaka Branch	Melaka	0	0	0	0	0	NULL	0.00

Figure 6: *Result of Query 5*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

branch_name	state	total_customers	total_accounts	total_deposits_myr	total_loans	total_loan_book_myr	total_outstanding_myr	ldr_pct	deposit_rank	customer_rank	loan_rank
Johor Bahru Branch	Johor	19	30	2,260,600.00	13	7,185,000.00	6,365,300.00	281.58	1	1	1
KLCC Branch	Kuala Lumpur	17	26	1,618,600.00	13	5,134,000.00	3,326,900.00	205.54	2	2	1
Petaling Jaya Branch	Selangor	11	19	708,300.00	8	2,320,000.00	1,551,000.00	218.98	3	3	3
Georgetown Branch	Penang	11	16	630,800.00	6	2,602,000.00	2,294,600.00	363.76	4	3	4
Shah Alam Branch	Selangor	8	12	531,500.00	4	875,000.00	658,000.00	123.80	5	5	5
Ipoh Branch	Perak	7	14	485,800.00	2	820,000.00	448,000.00	92.22	6	6	7
Subang Jaya Branch	Selangor	7	12	184,400.00	4	1,033,000.00	808,150.00	438.26	7	6	5
Kuching Branch	Sarawak	0	0	0.00	0	NULL	NULL	0.00	8	8	8
Kota Kinabalu Branch	Sabah	0	0	0.00	0	NULL	NULL	0.00	8	8	8
Melaka Branch	Melaka	0	0	0.00	0	NULL	NULL	0.00	8	8	8

Figure 7: *Result of Query 7*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

branch_name	state	total_deposits_myr	loans_outstanding_myr	ldr_pct	ldr_status
Subang Jaya Branch	Selangor	184,400.00	808,150.00	438.26	Over-lent ⚠
Georgetown Branch	Penang	630,800.00	2,294,600.00	363.76	Over-lent ⚠
Johor Bahru Branch	Johor	2,260,600.00	6,365,300.00	281.58	Over-lent ⚠
Petaling Jaya Branch	Selangor	708,300.00	1,551,000.00	218.98	Over-lent ⚠
KLCC Branch	Kuala Lumpur	1,618,600.00	3,326,900.00	205.54	Over-lent ⚠
Shah Alam Branch	Selangor	531,500.00	658,000.00	123.80	Over-lent ⚠
Ipoh Branch	Perak	485,800.00	448,000.00	92.22	Over-lent ⚠
Kuching Branch	Sarawak	0.00	NULL	0.00	No Loans
Kota Kinabalu Branch	Sabah	0.00	NULL	0.00	No Loans
Melaka Branch	Melaka	0.00	NULL	0.00	No Loans

Figure 8: *Result of Query 8*

7. Customer Wealth Segmentation (Q9)

Using the NTILE(4) window function, all 80 customers were divided into four wealth quartiles based on total account balance.

- **Premium (Q1):** Balances from MYR 145,000 to MYR 320,000. All T20 customers except two M40 outliers.

- **Affluent (Q2):** Balances from MYR 90,500 to MYR 143,500. Mixed T20 and M40 earners.
- **Mass Market (Q3):** Mid-range balances, predominantly salaried professionals.
- **Entry Level (Q4):** Balances below MYR 30,000. Younger customers and lower-income earners.

The presence of two M40 customers (Raj Gopal and Hazwan Arif) in the Premium segment is notable. It suggests that income tier alone is not sufficient to predict wealth accumulation and that savings behaviour matters more.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	full_name	income_tier	monthly_income	total_balance	wealth_quartile	customer_segment
8	Zulkifli Omar	T20	18500.00	320000.00	1	Premium
23	Noraini Hamid	T20	16800.00	275000.00	1	Premium
38	Wong Kok Wai	T20	18500.00	270000.00	1	Premium
68	Vikram Balaji	T20	13700.00	270000.00	1	Premium
76	Deepa Mohan	T20	11000.00	263000.00	1	Premium
58	Chan Wai Keng	T20	17600.00	240000.00	1	Premium
3	Abdullah Sani	T20	15200.00	235000.00	1	Premium
33	Aisyah Razak	T20	12200.00	228000.00	1	Premium
72	Kavitha Raj	T20	11600.00	225600.00	1	Premium
52	Lim Bee Geok	T20	11300.00	223000.00	1	Premium
64	Raj Gopal	M40	10200.00	202100.00	1	Premium
13	Shafiq Hamdan	T20	11800.00	195000.00	1	Premium
61	Rajan Krishnan	T20	15300.00	193000.00	1	Premium
48	Cheah Boon C...	T20	12900.00	182000.00	1	Premium
26	Nurul Izzati	T20	13900.00	168000.00	1	Premium
63	Muthu Pillai	T20	22500.00	155000.00	1	Premium
20	Hazwan Arif	M40	8800.00	151800.00	1	Premium
41	Ong Jian Wei	T20	14600.00	147000.00	1	Premium
55	Ng Sook Kuan	M40	9400.00	145600.00	1	Premium
43	Chan Wai Leong	T20	19800.00	145000.00	1	Premium
15	Ridhwan Kassim	M40	8900.00	143500.00	2	Affluent
17	Shahril Nizam	M40	10500.00	138000.00	2	Affluent
28	Azlina Mansor	T20	21000.00	130000.00	2	Affluent
6	Ismail Hashim	T20	12500.00	122000.00	2	Affluent
78	Meena Gopal	T20	16200.00	118000.00	2	Affluent
56	Ong Li Ping	T20	10800.00	115000.00	2	Affluent
32	Hazwani Ghazali	M40	9800.00	103700.00	2	Affluent
36	Tan Wei Liang	T20	11200.00	98800.00	2	Affluent
1	Ahmad Razali	M40	8200.00	90500.00	2	Affluent

Figure 9: *Result of Query 9*

8. Running Balance with Window Functions (Q10)

The running balance query for Account ID 1 (Ahmad Razali) uses SUM() OVER (PARTITION BY account_id ORDER BY transaction_date ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW). This computes a cumulative net cash flow per transaction in sequence. Ahmad receives a consistent MYR 5,200 salary credit monthly via online banking. After seven transactions including withdrawals, payments, and a transfer out,

the running net flow stands at MYR 11,470. This approach is directly applicable in overdraft detection and fraud monitoring systems.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

transaction_id	transaction_type	amount_myr	channel	description	transaction_date	balance_after_myr	running_net_flow_myr	txn_sequence
1	deposit	5,200.00	online	Salary Credit	2023-01-05 09:15:00	12,500.00	5,200.00	1
2	withdrawal	800.00	ATM	ATM Withdrawal	2023-01-10 14:22:00	11,700.00	4,400.00	2
3	payment	150.00	mobile	TNB Bill Payment	2023-01-15 11:30:00	11,550.00	4,250.00	3
4	deposit	5,200.00	online	Salary Credit	2023-02-05 09:10:00	16,750.00	9,450.00	4
5	transfer_out	3,000.00	online	IBG Transfer Out	2023-02-18 16:45:00	13,750.00	6,450.00	5
244	deposit	5,200.00	online	Salary Credit	2023-03-05 09:15:00	18,950.00	11,650.00	6
245	payment	180.00	mobile	Unifi TM Bill	2023-03-12 11:00:00	18,770.00	11,470.00	7

Figure 10: *Result of Query 10*

9. Customer 360 Portfolio View with CTEs (Q11)

Three CTEs were chained together to build a full financial snapshot per customer: one for accounts, one for loans, and one for cards. The final output joins these CTEs with the customers table via LEFT JOIN to ensure customers without loans or cards are still included. A net_financial_position column is computed as total deposits minus outstanding loans minus card balances. Most top depositors carry negative net positions due to large outstanding home loans. Wong Kok Wai is the strongest exception with a net position of MYR +252,800, because his home loan was fully settled before the analysis date.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	full_name	income_tier	monthly_income_myr	branch_name	num_accounts	total_deposits_myr	num_loans	total_borrowed_myr	outstanding_loans_myr	bad_loans	num_cards	credit_limit_myr	card_outstanding_myr	net_financial_position_myr
8	Zulkifli Omar	T20	18,500.00	Johor Bahru Branch	2	320,000.00	1	580,000.00	510,400.00	0	1	50,000.00	22,000.00	-212,400.00
23	Noraini Hamid	T20	16,800.00	KLCC Branch	2	275,000.00	1	520,000.00	468,000.00	0	1	40,000.00	18,500.00	-211,500.00
68	Vikram Balaji	T20	13,700.00	Johor Bahru Branch	2	270,000.00	1	720,000.00	684,000.00	0	1	50,000.00	23,400.00	-437,400.00
38	Wong Kok Wai	T20	18,500.00	KLCC Branch	2	270,000.00	1	600,000.00	0.00	0	1	40,000.00	17,200.00	252,800.00
76	Deepa Mohan	T20	11,000.00	Johor Bahru Branch	2	263,000.00	1	410,000.00	369,000.00	0	0	0.00	0.00	-106,000.00
58	Chan Wai Keng	T20	17,600.00	Georgetown Branch	2	240,000.00	1	650,000.00	617,500.00	0	1	45,000.00	20,200.00	-397,700.00
3	Abdullah Sani	T20	15,200.00	KLCC Branch	2	235,000.00	1	450,000.00	382,500.00	0	1	30,000.00	12,500.00	-160,000.00
33	Aisyah Razak	T20	12,200.00	KLCC Branch	2	228,000.00	1	380,000.00	323,000.00	0	1	35,000.00	15,600.00	-110,600.00
72	Kavitha Raj	T20	11,600.00	Johor Bahru Branch	3	225,600.00	1	350,000.00	315,000.00	0	0	0.00	0.00	-89,400.00
52	Lim Bee Geok	T20	11,300.00	Johor Bahru Branch	2	223,000.00	1	165,000.00	115,500.00	0	1	40,000.00	18,000.00	89,500.00
64	Raj Gopal	M40	10,200.00	Johor Bahru Branch	2	202,100.00	1	500,000.00	450,000.00	0	1	40,000.00	17,600.00	-265,500.00
13	Shafiq Hamdan	T20	11,800.00	KLCC Branch	2	195,000.00	1	148,000.00	118,400.00	0	1	30,000.00	13,600.00	63,000.00
61	Rajan Krishnan	T20	15,300.00	Petaling Jaya Branch	2	193,000.00	1	45,000.00	0.00	0	1	25,000.00	10,600.00	182,400.00
48	Cheah Boon C...	T20	12,900.00	Ipoh Branch	2	182,000.00	1	280,000.00	224,000.00	0	1	30,000.00	13,400.00	-55,400.00
26	Nurul Izzati	T20	13,900.00	Petaling Jaya Branch	2	168,000.00	1	80,000.00	0.00	0	1	25,000.00	10,200.00	157,800.00
63	Muthu Pillai	T20	22,500.00	Georgetown Branch	1	155,000.00	1	420,000.00	378,000.00	0	1	50,000.00	22,800.00	-245,800.00
20	Hazwan Arif	M40	8,800.00	Johor Bahru Branch	3	151,800.00	0	0.00	0.00	0	1	35,000.00	16,800.00	135,000.00
41	Ong Jian Wei	T20	14,600.00	Shah Alam Branch	2	147,000.00	1	35,000.00	22,750.00	0	1	20,000.00	7,800.00	116,450.00
55	Ng Sook Kuan	M40	9,400.00	Ipoh Branch	3	145,600.00	0	0.00	0.00	0	1	25,000.00	9,400.00	136,200.00
43	Chan Wai Leong	T20	19,800.00	Johor Bahru Branch	1	145,000.00	1	500,000.00	450,000.00	0	1	50,000.00	21,500.00	-326,500.00

Figure 11: *Result of Query 11*

10. Delinquency Detection (Q12)

The overdue loan query pulls from the vw_overdue_loans view and flags all missed and partial repayments where the due date has passed. All 17 flagged customers fall into the Serious Delinquency category, meaning their overdue periods exceed 90 days. The delinquency ranges from 843 days to 1,808 days.

The most overdue case is Loh Thim Fook, whose partial home loan payment has been unresolved for 1,808 days. The largest single payment shortfall is Azlina Mansor on an SME loan, owing MYR 9,266.56 for 843 days. A disproportionate number of delinquent customers are M40-tier borrowers, suggesting that middle-income earners face the most repayment

pressure. In practice, these accounts would be escalated to the credit remediation team for restructuring or legal recovery proceedings.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	loan_type	branch_name	income_tier	amount_due_myr	amount_paid_myr	shortfall_myr	due_date	paid_date	repayment_status	days_overdue	risk_flag
Loh Thim Fook	home	Subang Jaya Branch	M40	1,912.46	900.00	1,012.46	2021-04-14	NULL	partial	1808	Serious Delinquency
Roslan Bakar	personal	Subang Jaya Branch	M40	976.60	0.00	976.60	2022-03-03	NULL	missed	1485	Serious Delinquency
Ong Jian Wei	personal	Shah Alam Branch	T20	1,107.78	500.00	607.78	2022-07-01	NULL	partial	1365	Serious Delinquency
Deepa Mohan	home	Johor Bahru Branch	T20	2,230.18	1,000.00	1,230.18	2022-08-20	NULL	partial	1315	Serious Delinquency
Muthu Pillai	sme	Georgetown Branch	T20	6,885.16	3,000.00	3,885.16	2022-09-01	NULL	partial	1303	Serious Delinquency
Farhana Osman	personal	KLCC Branch	M40	635.98	0.00	635.98	2022-09-07	NULL	missed	1297	Serious Delinquency
Muzaffar Idris	personal	KLCC Branch	M40	1,091.49	600.00	491.49	2022-09-11	NULL	partial	1293	Serious Delinquency
Hazwani Ghazali	personal	Johor Bahru Branch	M40	1,540.96	800.00	740.96	2022-10-09	NULL	partial	1265	Serious Delinquency
Ong Li Ping	personal	Johor Bahru Branch	T20	1,212.96	0.00	1,212.96	2022-11-20	NULL	missed	1223	Serious Delinquency
Wong Mei Fong	personal	KLCC Branch	M40	882.55	0.00	882.55	2023-01-22	NULL	missed	1160	Serious Delinquency
Ng Boon Keat	personal	Johor Bahru Branch	M40	935.30	0.00	935.30	2023-02-13	NULL	missed	1138	Serious Delinquency
Ismail Hashim	personal	Shah Alam Branch	T20	1,001.14	0.00	1,001.14	2023-04-10	NULL	missed	1082	Serious Delinquency
Noraini Hamid	home	KLCC Branch	T20	2,830.43	1,500.00	1,330.43	2023-06-10	NULL	partial	1021	Serious Delinquency
Vikram Balaji	home	Johor Bahru Branch	T20	3,648.13	0.00	3,648.13	2023-07-01	NULL	missed	1000	Serious Delinquency
Abdullah Sani	home	KLCC Branch	T20	2,200.46	0.00	2,200.46	2023-09-15	NULL	missed	924	Serious Delinquency
Goh Pei Shan	personal	Petaling Jaya Branch	B40	689.46	300.00	389.46	2023-10-13	NULL	partial	896	Serious Delinquency
Azlina Mansor	sme	Johor Bahru Branch	T20	9,266.56	0.00	9,266.56	2023-12-05	NULL	missed	843	Serious Delinquency

Figure 12: *Result of Query 12*

11. Card Spending Insights (Q6 and Q14)

Travel generates the highest average credit card spend per transaction at MYR 627.29, driven by Malaysia Airlines (MYR 2,250 average per booking) and AirAsia. Groceries are the most frequent category at 20.41% of all transactions, but with a lower average of MYR 263.67 per visit. Online shopping via Shopee MY, Lazada MY, and Zalora accounts for MYR 5,078 in total Shopee spend alone, confirming Malaysian e-commerce dominance. Healthcare is the lowest frequency category but commands one of the higher per-transaction averages (MYR 214.50), reflecting private hospital charges.

The top merchant ranking query uses RANK() OVER (PARTITION BY merchant_category ORDER BY SUM(amount) DESC) to rank merchants within each spending category. This is directly useful for designing co-branded rewards programmes and negotiating merchant partnerships.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

merchant_category	total_transactions	total_spend_myr	avg_spend_myr	min_spend_myr	max_spend_myr	pct_of_transactions
travel	7	4,391.00	627.29	55.00	1,800.00	14.29
online	8	3,900.00	487.50	295.00	840.00	16.33
groceries	10	2,636.70	263.67	65.80	345.60	20.41
utilities	8	1,423.00	177.88	158.00	220.00	16.33
dining	9	738.90	82.10	33.50	136.00	18.37
healthcare	3	643.50	214.50	95.50	420.00	6.12
petrol	2	268.00	134.00	120.00	148.00	4.08
entertainment	2	90.00	45.00	35.00	55.00	4.08

Figure 13: *Result of Query 6*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

merchant_category	spend_rank	merchant_name	txn_count	total_spend_myr	avg_spend_myr
groceries	1	Aeon	9	2,911.20	323.47
groceries	2	Jaya Grocer	11	2,552.80	232.07
groceries	3	Tesco Malaysia	6	2,122.60	353.77
dining	1	Nando's Malaysia	8	912.00	114.00
dining	2	Secret Recipe	9	887.00	98.56
dining	3	KFC	6	308.70	51.45
travel	1	Malaysia Airlines	6	13,500.00	2,250.00
travel	2	AirAsia	8	6,700.00	837.50
travel	3	Grab	9	558.50	62.06
online	1	Shopee MY	10	5,078.00	507.80
online	2	Lazada MY	7	3,478.00	496.86
online	3	Zalora MY	5	1,928.00	385.60
utilities	1	TNB	6	1,191.00	198.50
utilities	2	Unifi TM	4	880.00	220.00
utilities	3	Maxis	5	860.00	172.00
entertainment	1	Sunway Lagoon	2	400.00	200.00
entertainment	2	Genting Highlands	1	350.00	350.00
entertainment	3	Netflix MY	5	275.00	55.00
petrol	1	Shell Malaysia	6	875.00	145.83
petrol	2	Petronas	6	834.00	139.00
petrol	3	BHPetrol	4	515.00	128.75
healthcare	1	Pantai Hospital	6	3,390.00	565.00
healthcare	2	KPJ Hospital	6	2,640.00	440.00
healthcare	3	Caring Pharmacy	6	720.50	120.08

Figure 14*: Result of Query 14*

11. Stored Procedure Output (Bonus Query)

Calling GetCustomerPortfolio(1) returns Ahmad Razali's full financial profile in one database call. He holds three active accounts (savings, fixed deposit, current) with a combined balance of MYR 90,500. He has one active personal loan with MYR 17,500 outstanding at MYR 771.86 monthly. He holds two active cards: a Visa debit and a Mastercard credit card with MYR 4,200 outstanding against a MYR 15,000 limit. This stored procedure demonstrates how a bank's customer service interface could retrieve a complete client profile without running multiple separate queries.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	full_name	nric	income_tier	monthly_income	customer_since	home_branch
1	Ahmad Razali	850312-10-1007	M40	8200.00	2018-05-14	KLCC Branch

Figure 15: *Result of Query 16*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

account_number	account_type	balance	status
MY-482910-23	savings	12500.00	active
MY-731824-11	fixed_deposit	50000.00	active
MY-374829-88	current	28000.00	active

Figure 16: *Result of Query 16*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

loan_type	principal	outstanding	monthly_instalment	maturity_date	status
personal	25000.00	17500.00	771.86	2025-05-14	active

Figure 17: *Result of Query 17*

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

card_type	card_network	card_number	credit_limit	outstanding_balance	status	expiry_date
debit	Visa	****4821	NULL	0.00	active	2024-01-01
credit	Mastercard	****9374	15000.00	4200.00	active	2024-06-01

Figure 18: *Result of Query 18*

Key Takeaways and Recommendations

Immediate actions required:

- All seven active branches exceed the BNM LDR guideline. Deposit mobilisation campaigns and tighter loan approval controls are needed, particularly at Subang Jaya (438%) and Georgetown (364%).
- Seventeen loan accounts are in serious delinquency. These should be reviewed for formal NPL classification and referred for credit remediation.

Growth opportunities:

- The Premium segment holds significant cross-selling potential for fixed deposits, unit trusts, and private banking services.
- Travel and online card spending categories show the highest average ticket sizes. A co-branded travel rewards credit card with Malaysia Airlines or AirAsia would be commercially viable.

- The three inactive East Malaysian branches (Kuching, Kota Kinabalu, Melaka) represent untapped market territory.

Conclusion

This project demonstrates end-to-end capability in relational database design, synthetic data generation, and business analytics using SQL. The entire database, from schema to data to analysis queries, was built from scratch without any external datasets or automated tools. The 14 analysis queries collectively cover branch performance benchmarking, customer wealth segmentation, credit risk monitoring, transaction trend analysis, and card spending intelligence.

Advanced SQL features including CTEs, six types of window functions, stored procedures, parameterised views, and complex multi-table joins were used throughout. The findings are grounded in realistic Malaysian banking context, referencing BNM regulatory guidelines, Malaysian income tier classifications, and locally relevant merchant and financial product data. This project serves as a practical demonstration that SQL, when applied thoughtfully to a well-designed schema, is a powerful standalone business intelligence tool.